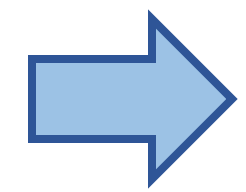
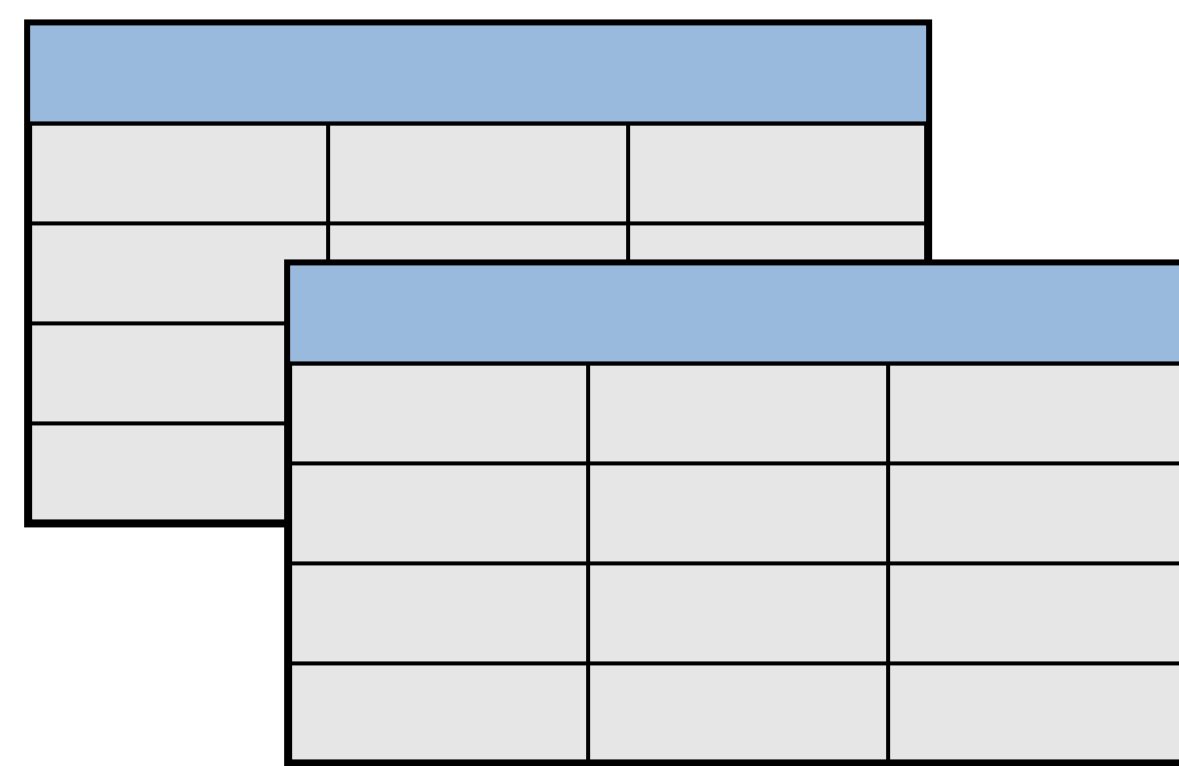


Data Input & Era Partition

TV Competition Shows
(34 Seasons of DWTS)

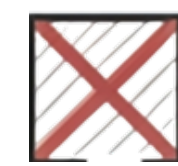
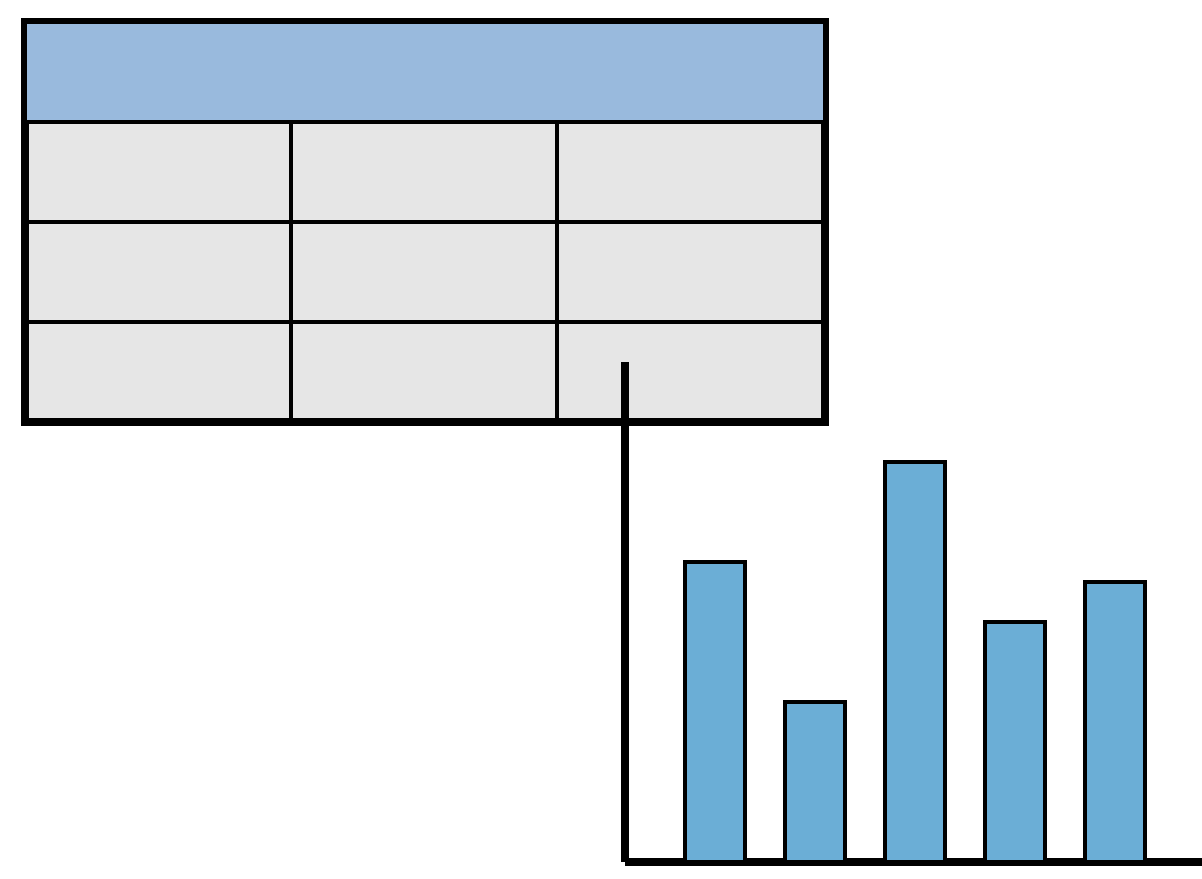


1. Rank-Based Era



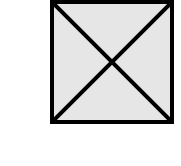
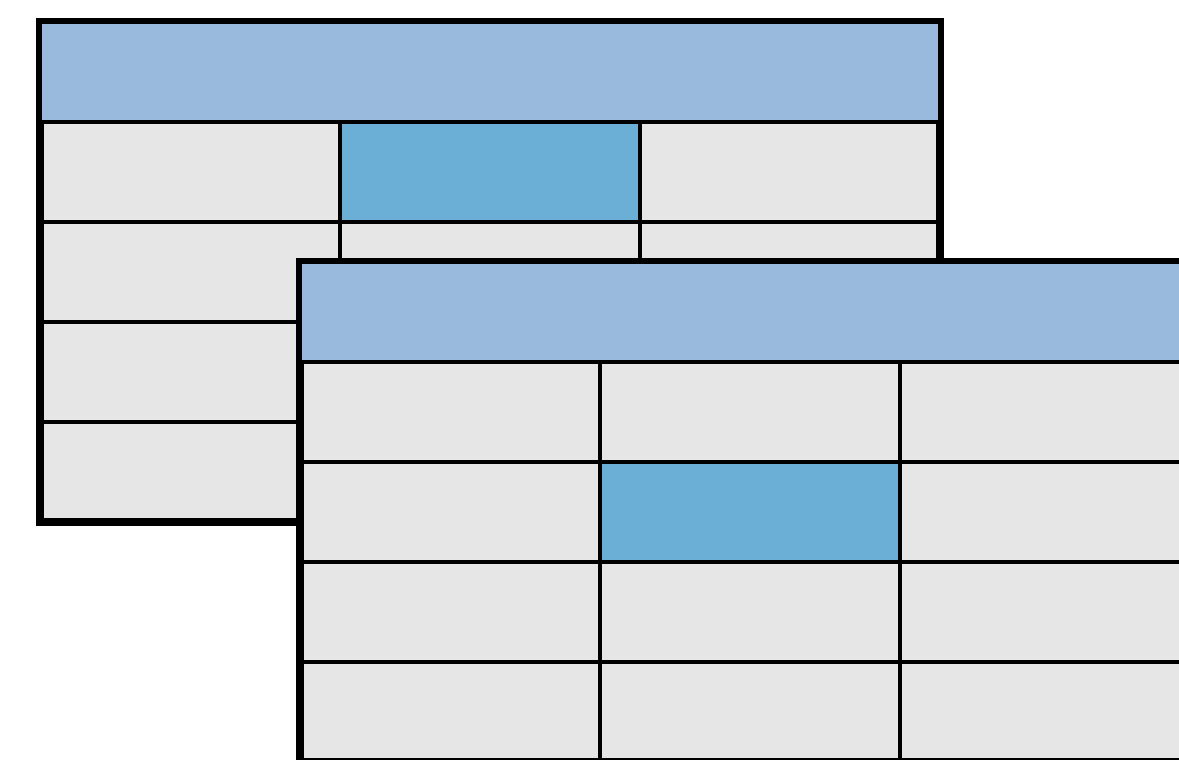
Historical Data Partitioned by Rules

2. Percentage-Based Era



Missing Fan Vote Data

3. Bottom-2 Rules Era



Missing Fan Vote Data

Module 1: Bayesian Estimation Model

Judge Scores

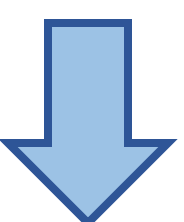
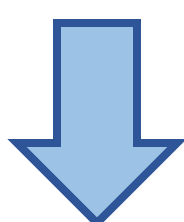


image sensor

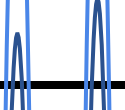
Elimination
Outcomes

A. Model Components

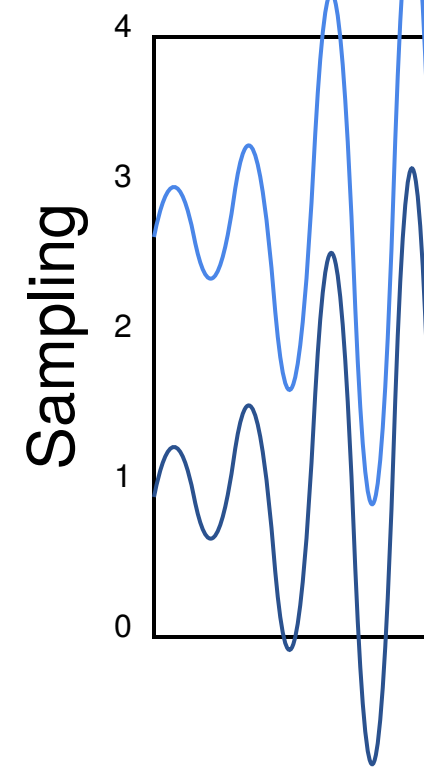
Prior: Dirichlet Distribution
(Weakly Info/Non-Info)



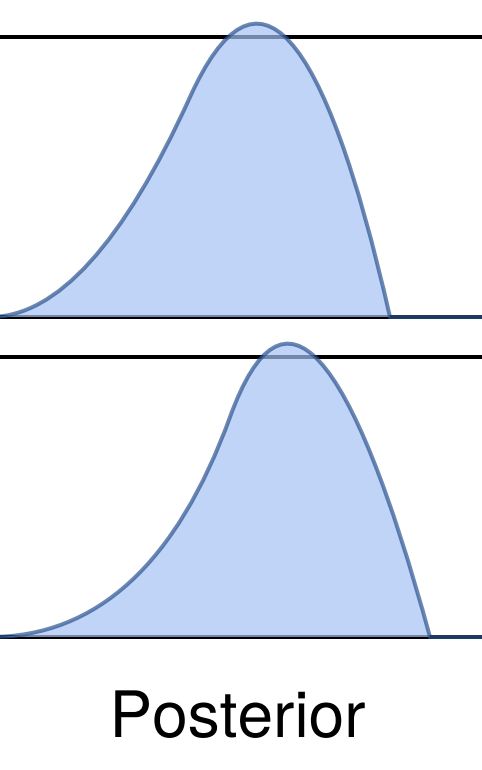
Likelihood:
Constraint-Based (from
outcomes)



B. MCMC Sampling

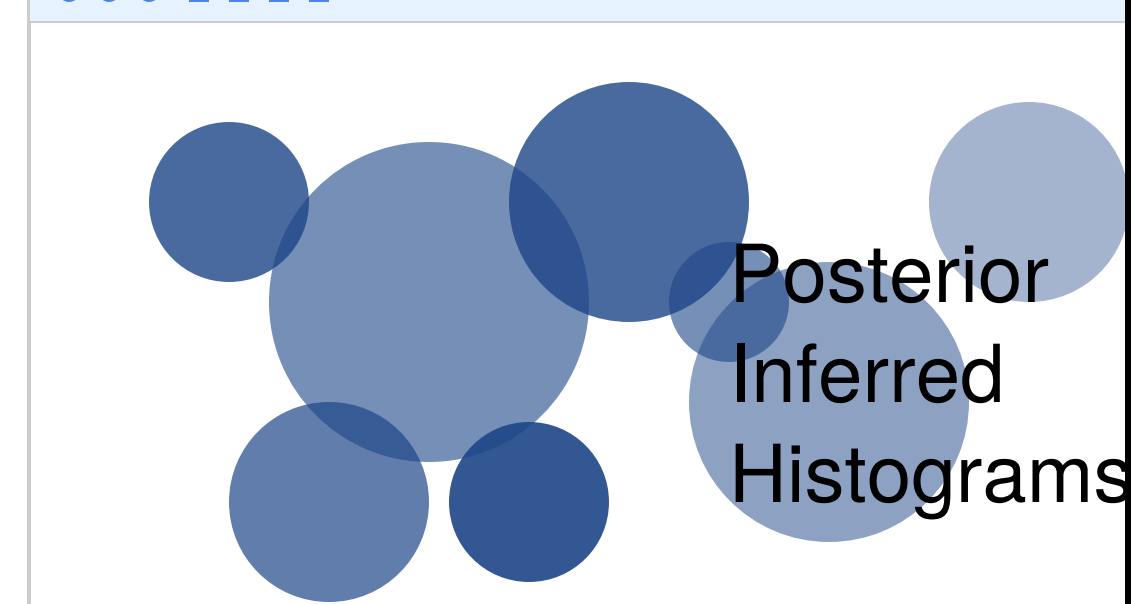


Chain



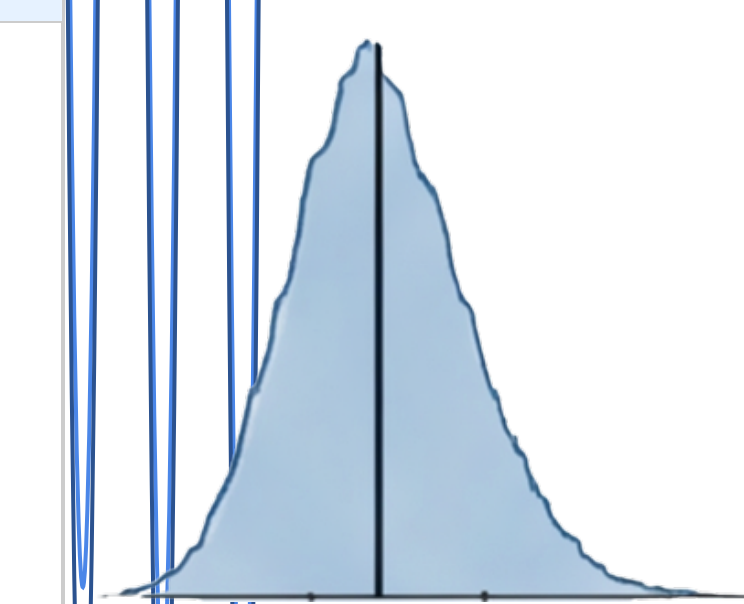
Posterior

DZI viewer



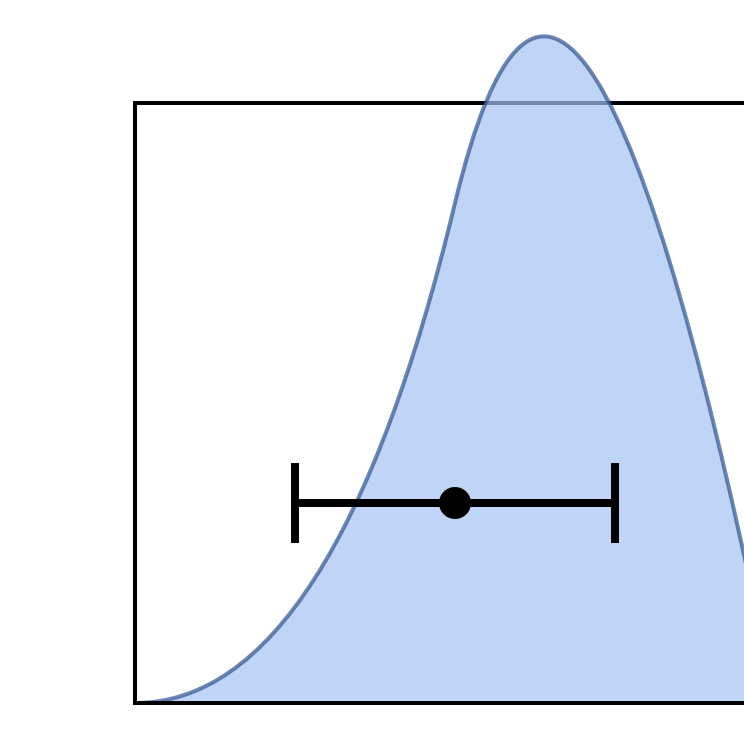
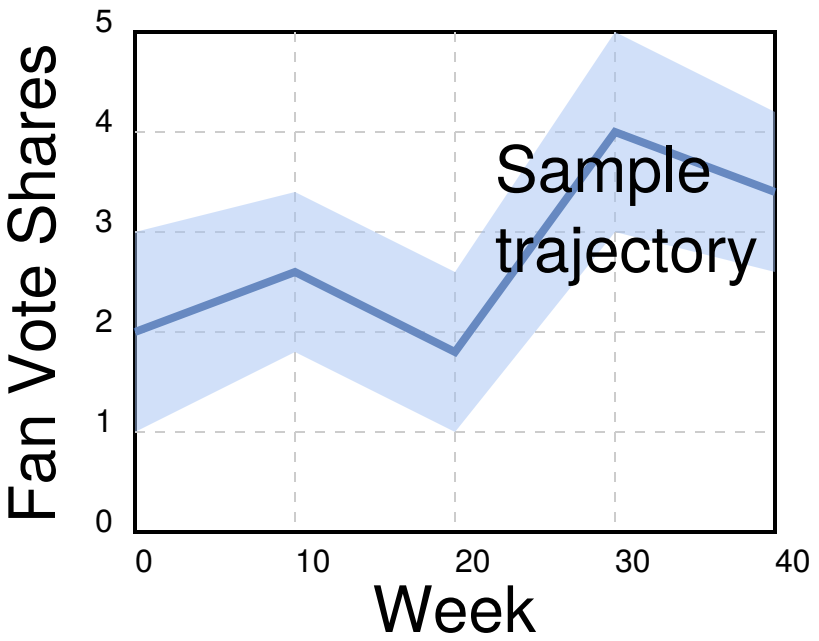
Real-time whole-slide
3D image review

Posterior
Inferred
Histograms



Quantified
Uncertainty
(95% HDI)

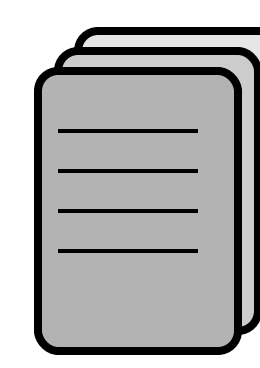
Posterior Inferred
Weekly Fan
Vote Shares



Quantified
Uncertainty
(95% HDI)

Module 2: Counterfactual Analysis & Mechanism Evaluation

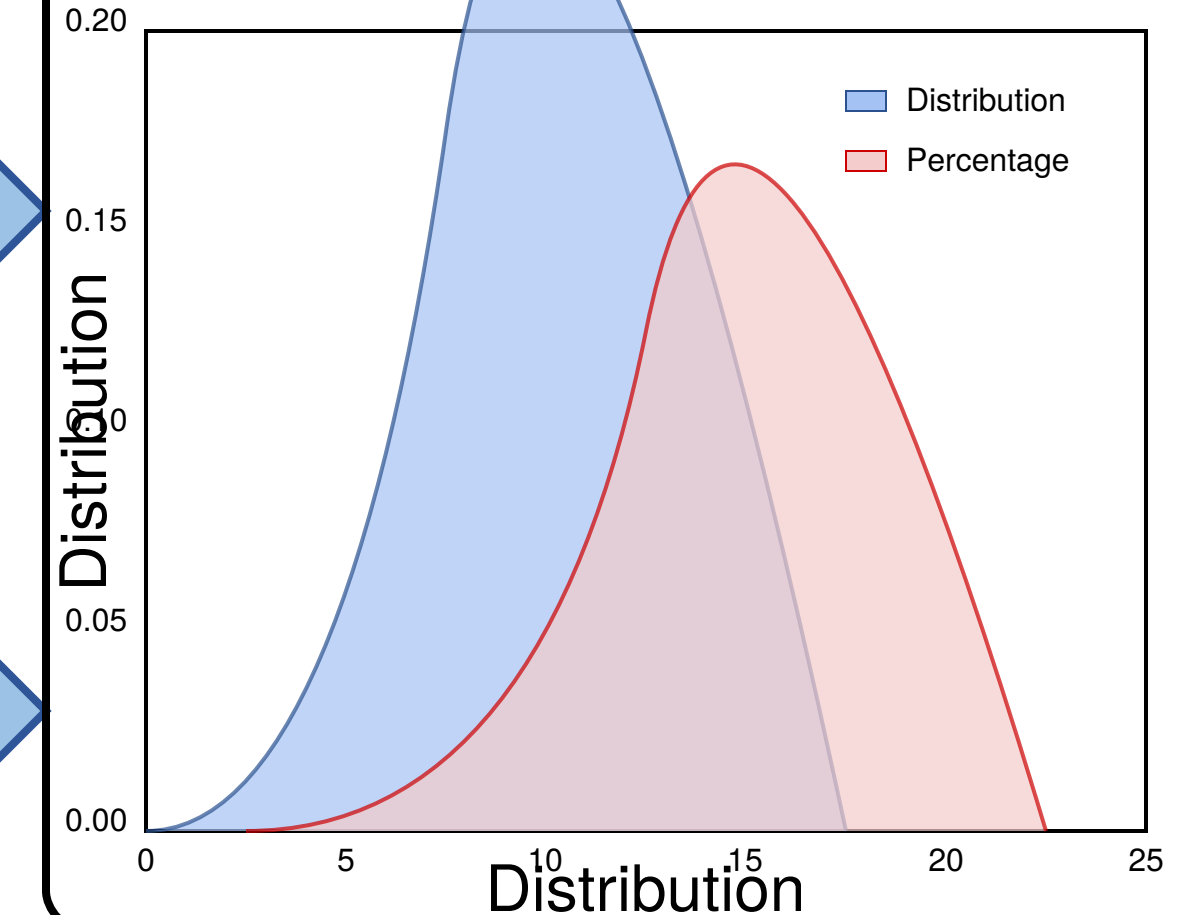
Historical
Rules



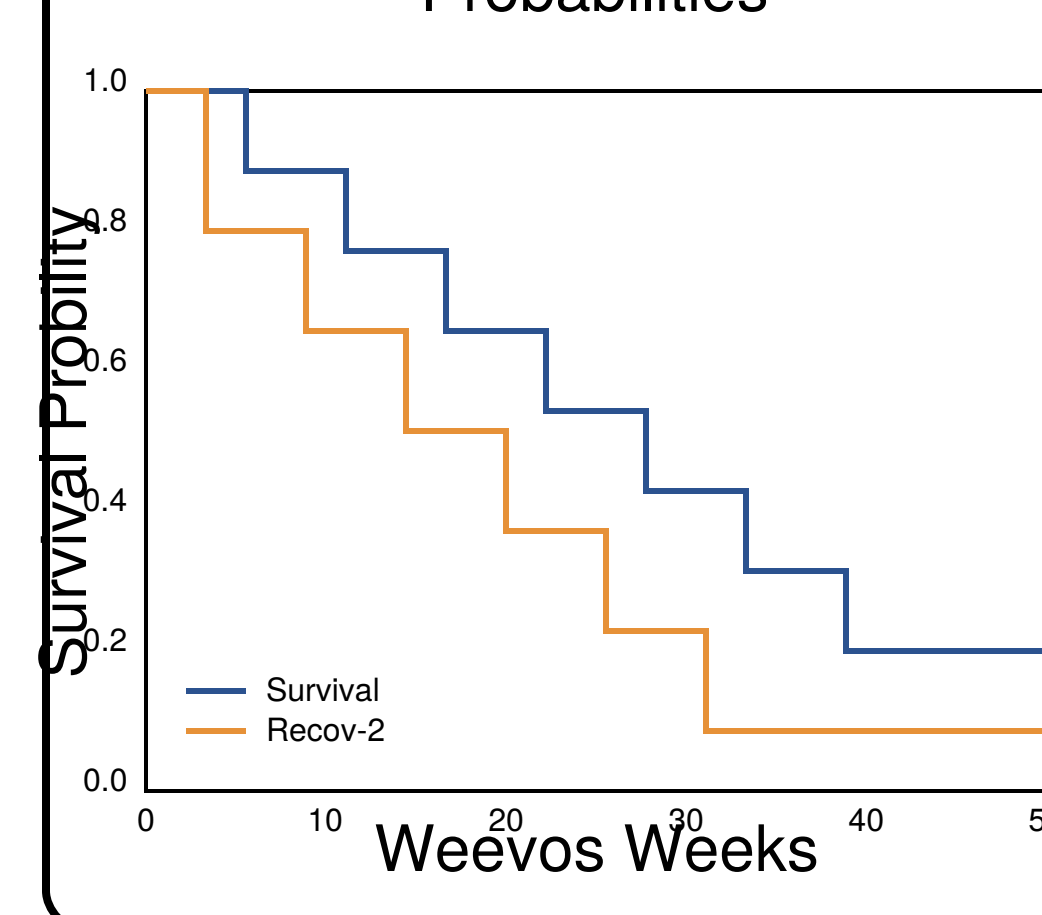
Inferred
Vote Shares



Simulation of Outcomes
(Rank vs. Percentage)



Assess Bottom-2
Impact on Survival
Probabilities



Characterized Systemic Rule Divergence

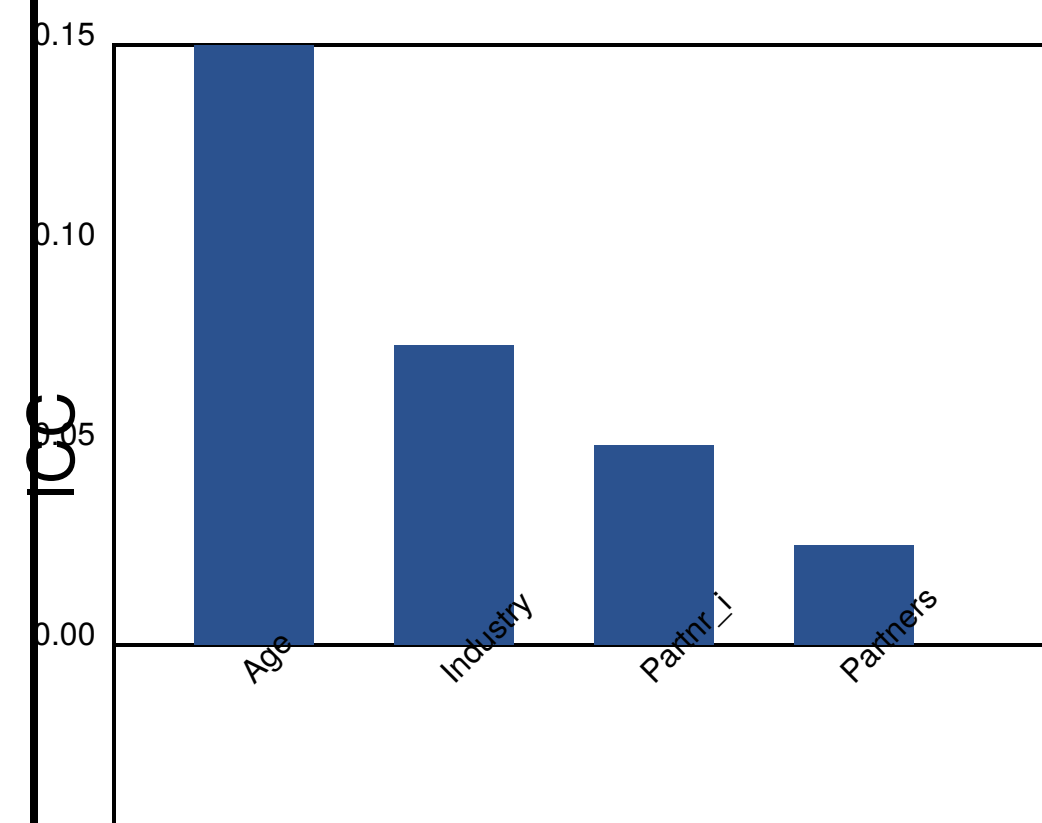
Module 3: Linear Mixed-Effects (LME) & Non-linear Attribution

Inferred Vote
Shares &
Judge
Scores

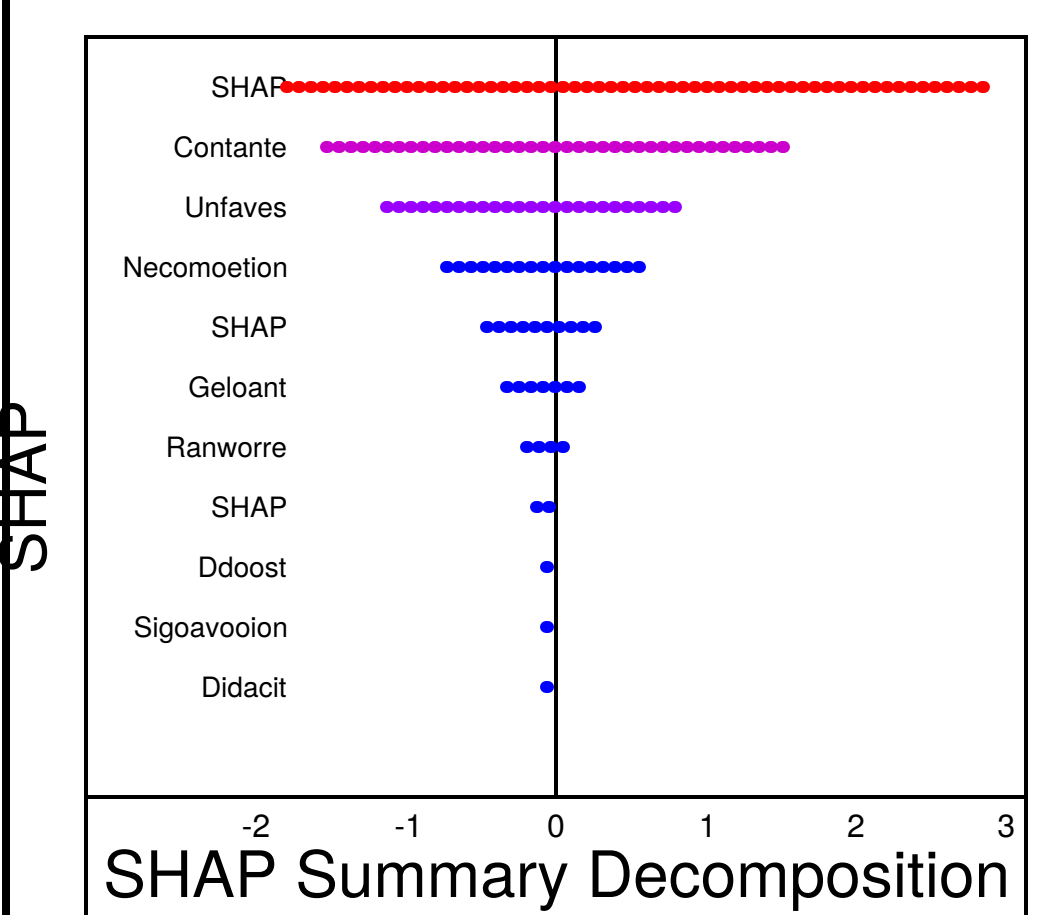
Contestant
Features

Partner
Identity

LME Model
(Fixed: Age, Industry;
Random: Partners)



XGBoost & SHAP
Decomposition

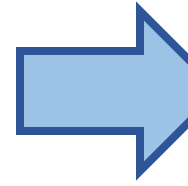


Separated Fixed/Random Effects and
Feature Importance Comparison

Separated Fix//Random Effects and
Feature Importance Comparison

Module 4: Dynamic Voting Mechanism Design

Fairness,
Engagement,
Unfair
Eliminations
Criteria

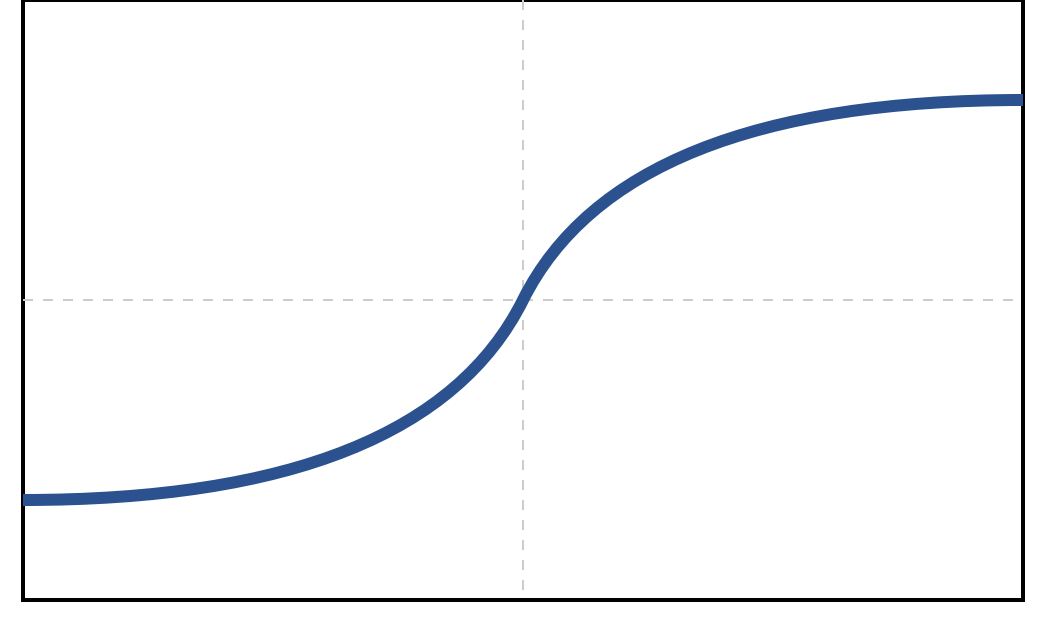


Static
Optimization
(Grid Search for
Optimal Weight)

(Grid Search for
Optimal Weight)

Dynamic Sigmoid Controller
(Judge Weight evolution)

Weight

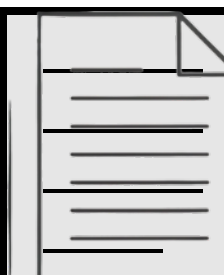


Optimized Voting Mechanism
with Sensitivity Analysis

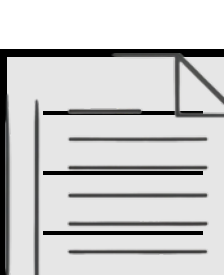
Method
Application
&
Optimization

Nucleus detection

Deliverables



Memorandum to
Executive Producers
(Deemphasized)



Competition Format
Recommendations
Evolutionary Format
with Dynamic Weight